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Myelomeningocele and Other Spinal Dysraphisms

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SUMMARY PEARLS

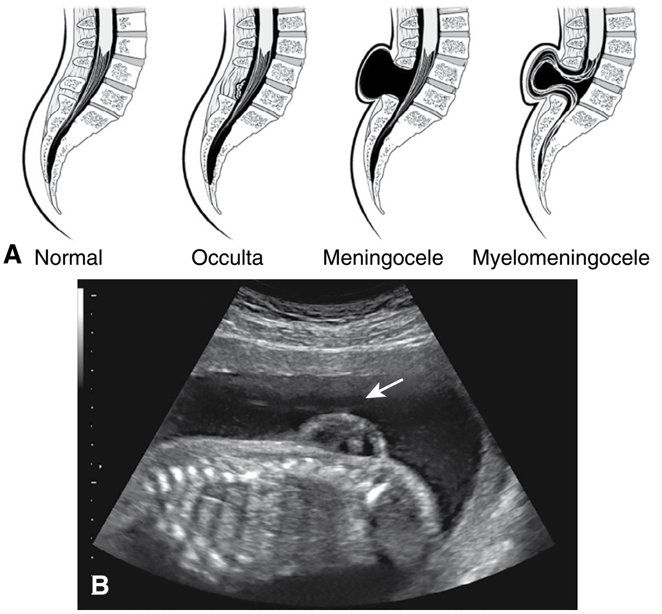
- Myelomeningocele (MMC) is the second most common disabling condition in the pediatric population.
- The embryonic development of the brain and spinal cord is complex, and many potential insults or abnormalities during the first 4 weeks of gestation may lead to spinal dysraphism.
- Folic acid supplementation of 400 µg daily has been shown to significantly decrease the prevalence and severity of neural tube defects (NTDs), and 4 mg of folate daily for females who have delivered a child with a NTD decreases the risk for a future pregnancy complicated by NTD.
- Fetal ultrasound is the gold standard for prenatal diagnosis of MMC and is performed between 18 and 22 weeks of gestation.
- Fetal surgery performed by 26 weeks of gestation has resulted in improved motor function, self-care skills, and bladder function, but it carries risks to the fetus and mother.
- Latex sensitivity is exceedingly common in the MMC population, and products containing latex, including gloves and catheters, must be avoided.
- Hydrocephalus and shunt function require ongoing close surveillance and are associated with dysregulation of the hypothalamic-pituitary axis, which may result in growth hormone deficiency and short stature as well as precocious puberty.
- Orthopedic deformities may be congenital or acquired, and while early treatment of clubfoot deformities is recommended, hip surveillance is no longer recommended, and surgical intervention for hip dislocation has not demonstrated any improvement in ambulatory ability.
- The functional level of a child with MMC is determined primarily by motor level. Assistive equipment and bracing should be utilized to improve function.
- Individuals with MMC have better verbal skills than nonverbal skills, which may lead to an overestimation of their cognitive abilities, and neuropsychology testing is highly recommended to assist with school interventions and planning for adulthood.

Epidemiology

Spinal dysraphism is the abnormal differentiation and/or incomplete closure of the neural tube during gestation (typically between 24 and 26 days of gestation).[92] It is estimated that the global prevalence of neural tube defects (NTDs) is 20 per 10,000 births.[57] However, this may be an underestimate because many areas of the world do not have a robust surveillance system.[120] There is significant disparity worldwide, with a higher prevalence and adverse outcomes reported in developing countries. Countries who have implemented mandated folic acid fortification in staple foods, such as wheat flour, rice, and cereal grains, have reported decreases of 19% to 55% in the prevalence of NTDs to a rate of 5 to 6 per 10,000 live births.[25] In the United States, the Centers for Disease Control and Prevention reports that spina bifida occurs in 1 of every 2875 births or a prevalence of 3.5 per 10,000 births.[103] However, there are some striking differences among the population. American Indians and Alaskan Natives have the highest prevalence (4 per 10,000 births) followed by Hispanics (3.8 per 10,000 births). The non-Hispanic Black population has the lowest prevalence (2.73 per 10,000). The non-Hispanic White population has a prevalence of 3.09 per 10,000. In 1988, the US Food and Drug Administration required that all grain products that are labeled as "enriched" have folic acid added. Studies demonstrated a 28% reduction in the prevalence of NTDs after this fortification initiative[116] and a 70% decrease in the severity of the lesion, mostly due to a decrease in the number of lesions occurring in the upper spine.[72]

Classification and Nomenclature

Spina bifida has been used to describe multiple conditions associated with a failure of the vertebral arches to close. It has been used nonspecifically to describe all spinal dysraphisms, although spina bifida refers to the fusion failure of the spinal bony elements only and not necessarily the soft tissue.[61] Anatomic classification is most commonly used. Lesions are first categorized as open or closed. In open spinal dysraphism, the nervous tissue is exposed



• **Fig. 49.1** Spina bifida. (A) Spina bifida is a birth defect of the spinal cord caused when the neural tube does not completely close, but the rest of development continues. The result is the emergence of meninges and neural tissue through the vertebral column. (B) Fetal myelomeningocele is evident in this ultrasound taken at 21 weeks. (From Betts JG, Young KA, Wise JA, et al. *Anatomy and Physiology*. OpenStax; 2013.)

to the outside, and there may be leakage of spinal fluid. However, in closed spinal dysraphisms, there is full epithelialization of the neural tissue. Cutaneous birth marks, including a hairy tuft and/or dorsal dimple, are present in up to 50% of closed dysraphisms.[31,113] Closed spinal dysraphisms are further characterized by the presence or absence of a subcutaneous mass.

Fig. 49.1 provides a visual representation of normal spine anatomy vs. closed and open spinal dysraphisms. The absent vertebrae are noted in the occulta lesion, with the spinal cord and sheath remaining within the spinal canal. Skin covers this defect. The meningocele represents a closed dysraphism with a subcutaneous mass. In this case, the subcutaneous mass is a sac containing the meninges and spinal fluid but no neural tissue. This is also covered with skin, although there have been reports of open meningoceles. There is often no significant neural injury with either the meningocele or the occulta type lesions. However, tethered cord syndrome may develop as the child grows. Some closed spinal dysraphisms, such as spinal lipomas, tight filum terminale, diastematomyelia, and caudal agenesis, are associated with tethered cord syndrome, and tethered cord syndrome is often the presenting symptom. The final diagram of myelomeningocele (MMC) demonstrates the herniation of the neural tissue through the vertebral and skin defects. This results in significant neural deficits in sensation, motor function, bowel and bladder function, possible Chiari II malformation, and cognitive and psychological impairments.

Older terminology for open and closed spinal dysraphisms is still found in the literature and in practice. Open spinal dysraphisms are also known as spina bifida aperta or spina bifida cystica. Spina bifida occulta refers to closed spinal dysraphisms. Box 49.1 lists the cliniconeuroradiologic classification of spinal dysraphisms.

A separate classification system relies on the embryologic stage at which the abnormality occurred. In this system, the NTD is either due to a disorder of primary neurulation or of secondary neurulation. Primary neurulation defects include craniorachischisis, anencephaly, and open spinal dysraphisms (MMC and myelocele). Craniorachischisis is a rare and fatal NTD with both anencephaly and an open spinal dysraphism from the cervical to the lumbar level. Meningoceles, though most often skin covered, are also included as a disorder of primary neurulation. Secondary neurulation defects may result in caudal agenesis, caudal duplication syndrome, segmental spinal dysgenesis (SSD), and lumbosacral or caudal lipoma.[119] A junctional NTD is described as a functional disconnection between the upper primary spinal cord and the lower secondary spinal cord with the two areas connected by a thin fibrous band and appears to correspond to the bony anomalies seen in SSD.[119]

• BOX 49.1 Cliniconeuroradiologic Classification of Spinal Dysraphism

Open Spinal Dysraphism

- Myelomeningocele
- Myelocele
- Hemimyelomeningocele
- Hemimyelocele

Closed Spinal Dysraphism

With a Subcutaneous Mass

- Lumbosacral
    - Lipoma with dural defect
    - Lipomyelomeningocele
    - Lipomyeloschisis
- Terminal myelocystocele
- Meningocele

Cervical

- Cervical myelocystocele
- Cervical myelomeningocele
- Meningocele

Without a Subcutaneous Mass

Simple Dysraphic States

- Posterior spinal bifida
- Intradural and intramedullary lipoma
- Filum terminale lipoma
- Tight filum terminale
- Abnormally long spinal cord
- Persistent terminal ventricle

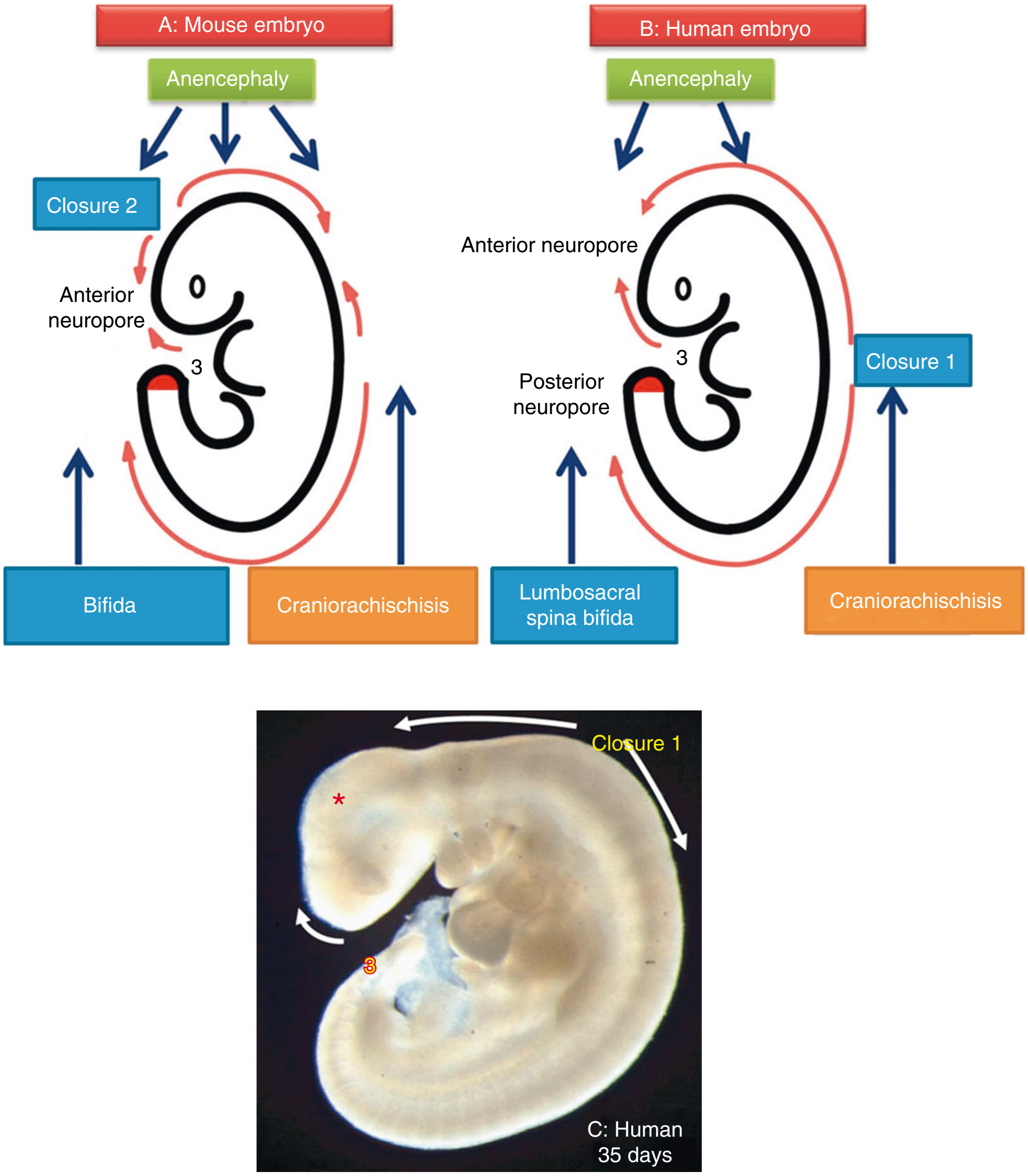
Complex Dysraphic States

- Dorsal enteric fistula
- Neurenteric cysts
- Split cord malformations (diastematomyelia and diplomyelia)
- Dermal sinus
- Caudal regression syndrome
- Segmental spinal dysgenesis

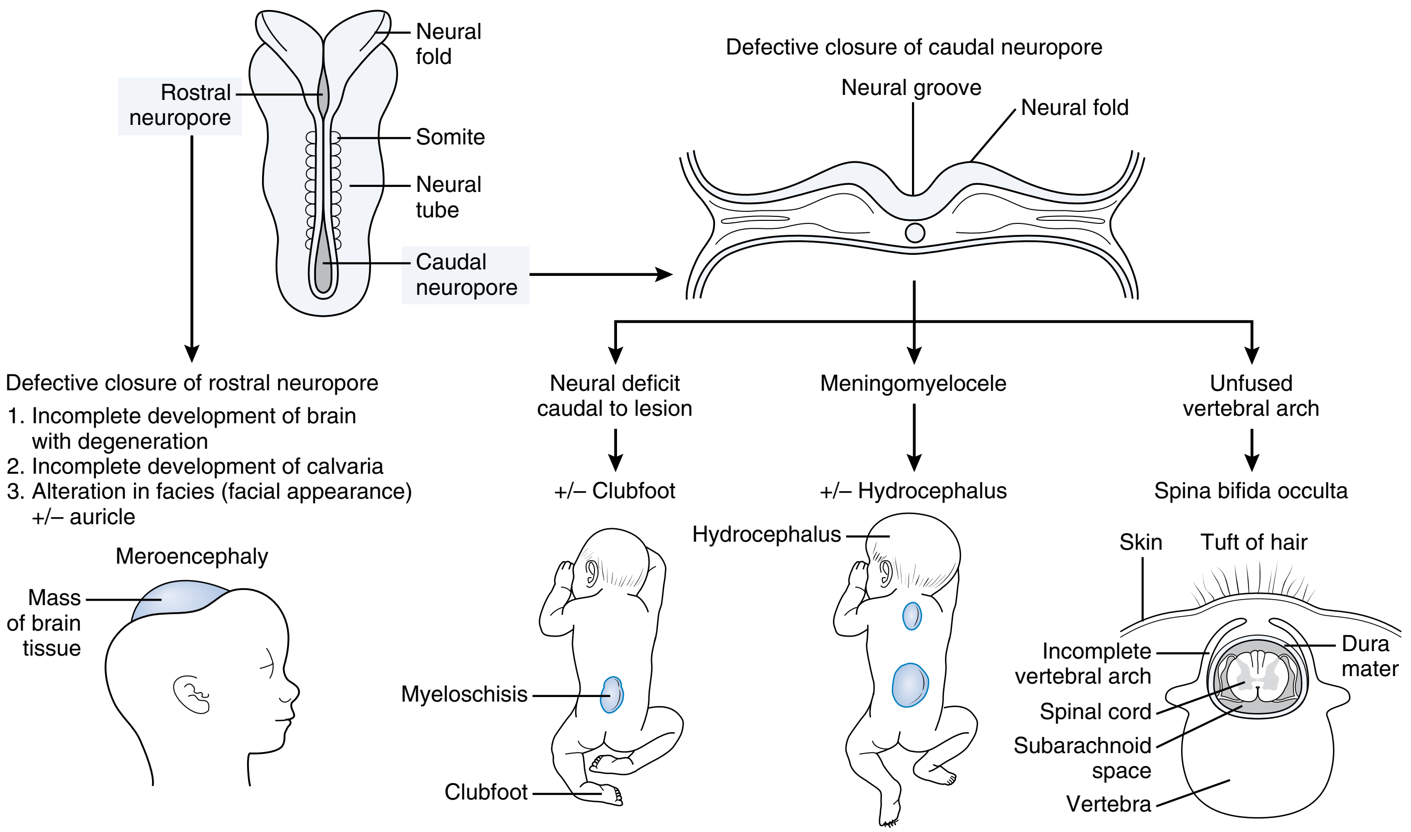
Embryology and Pathophysiology

The normal development of the spinal cord is important to understand when evaluating children with spinal dysraphisms. The child with MMC is different compared to the child who sustains a traumatic spinal cord injury. A comprehensive approach from the medical team with an understanding of the pathophysiology of spinal dysraphism and comorbidities is strongly recommended. The spinal cord is formed through the gastrulation, primary neurulation, and secondary neurulation phases. During weeks 2 and 3 of gestation, the embryonic disc becomes trilaminar with the addition of the mesoderm layer between the ectoderm and endoderm layers. The mesoderm is formed by the migration of epiblastic cells through the Henson node to the epiblast-hypoblast interface. The notochord is formed with epiblastic cell migration along the midline and is the structure around which the vertebrae form. The notochord is the inductor of the neural ectoderm.[92] Studies have theorized that persistent midline adhesions of the endoderm and ectoderm could cause a deviation in the epiblast migration leading to a splitting or malformation of the notochord.[15] This may induce the formation of a split spinal cord or paired hemicords. Vertebral anomalies may also occur, including hemivertebrae, butterfly vertebrae, and block vertebrae.[77] Split cord malformations may cause paraplegia, gait abnormalities, leg length discrepancy, and foot changes.[61]

Primary neurulation occurs in weeks 3 and 4 of gestation. The developing notochord induces the ectoderm to form the neuroectoderm. This flat neural ectoderm plate, or neural plate, invaginates at the midline to form the neural groove and neural folds. It progressively bends until the neural folds fuse forming the neural tube. This fusion proceeds bidirectionally in waves of closure with the rostral neuropore closing at day 25 and the caudal neuropore closing at day 27 or 28[28,92] (Fig. 49.2). The neural tube dissociates



• **Fig. 49.2** Diagrammatic representation of the main events of neural tube closure in mouse (A) and human (B) embryos. The main types of neural tube defect resulting from failure of closure at different levels of the body axis are indicated. The *red shading* on the tail bud indicates the site of secondary neurulation in both species. (C) Human embryo aged 35 days postfertilization from the Human Developmental Biology Resource (www.hdbr.org). Neurulation has recently been completed in the low spinal region. The positions of closures 1 and 3 and the directions of closure are marked. The midbrain in this human embryo *(red asterisk)* is relatively small compared with the corresponding stage of mouse development. This may have rendered closure 2 an unnecessary intermediate step in achieving cranial neural tube closure in humans. (From Copp AJ. Neurulation in the cranial region—normal and abnormal. *J Anat.* 2005;207:623-635.)



• **Fig. 49.3** Schematic illustrations showing the embryologic basis of neural tube defects. Meroencephaly, partial absence of brain, results from defective closure of the rostral neuropore, and meningomyelocele results from defective closure of the caudal neuropore. (Redrawn from Moore KL, Persaud TVN, Torchia MG, eds. Nervous system. In: *The Developing Human: Clinically Oriented Embryology.* 9th ed. Elsevier Saunders; 2013.)

from the ectoderm and will form the brain and spinal cord. The overlying ectoderm becomes the epidermis. It is thought that most of the spinal cord to S2 is formed during primary neurulation.[28] A failure in fusion can result in cranial defects, anencephaly, meroencephaly, or open spinal dysraphism (Fig. 49.3). This is a complex process, and the cause for failure of fusion appears to be multifactorial. Lack of folate and specifically the inability to provide methyl donors for methylation reactions have been shown to cause a failure in fusion, though the exact mechanism remains unknown.[14] Recent research has investigated the balance of cell death vs. proliferation in neural tissue development and neural tube closure. Both genetic and environmental factors can influence this balance. In the mouse model, it has been demonstrated that mice with spina bifida have an altered balance compared to mice without spina bifida.[9]

Secondary neurulation commences with the closure of the caudal neuropore. This process is species specific, and secondary neurulation in humans may resemble the process seen in mouse or chicken embryos. In the mouse embryo, a mass of pluripotent cells forms the caudal cell mass (CCM) at the caudal end of the neural plate (see Fig. 49.3). Cavitation forms a central lumen, and additional cells from the CCM are recruited to form a secondary neural tube. This neural tube is directly continuous with the primary neural tube and has a single lumen.[28,61] In the chick embryo, a medullary cord is formed from the cells of the CCM. The medullary cord consists of outer and inner cell groups. With cavitation between the two groups, multiple tubules are formed. These tubules will eventually coalesce to form a single lumen and the secondary neural tube. Of note, this secondary neural tube is not initially connected to the primary neural tube. It will eventually fuse with the primary neural tube with an overlap zone in which the primary neural tube is located dorsally and the secondary neural tube is located ventrally.[28] A disruption in the fusion between the primary and secondary neural tubes may lead to segmentation defects. There is continued debate regarding which process most resembles secondary neurulation in humans. This secondary neural tube forms the tip of the conus medullaris, the filum terminale, and the cauda equina.[61] With closure of the cranial neuropore and occlusion of the caudal neurotube, there is a closed fluid-filled space subjected to intraluminal pressures driving expansion of the brain and ventricular system.[61] It is hypothesized that an open spinal defect with leakage of cerebrospinal fluid (CSF) in utero results in decreased formative pressures. It prevents the expansion of the ventricular system and in turn affects brain formation leading to a small posterior fossa and herniation of the cerebellar vermis and medulla into the spinal canal (Chiari II malformation).[55] The Chiari II malformation can obstruct the flow of CSF leading to hydrocephalus and possibly cystic dilation of the brainstem and upper cervical spinal cord (syringomyelia).[55] Unfortunately, experimental evidence has not yet verified this hypothesis.

Risk Factors

There are many risk factors associated with the development of a NTD, including genetic and environmental factors. Maternal diabetes, possibly hyperthermia, anticonvulsant agents, obesity, and toxins have all been implicated as well as genetic influences.[55] While no single gene has been identified as a cause for NTDs, there is considerable support for genetic influence in the development of NTDs.

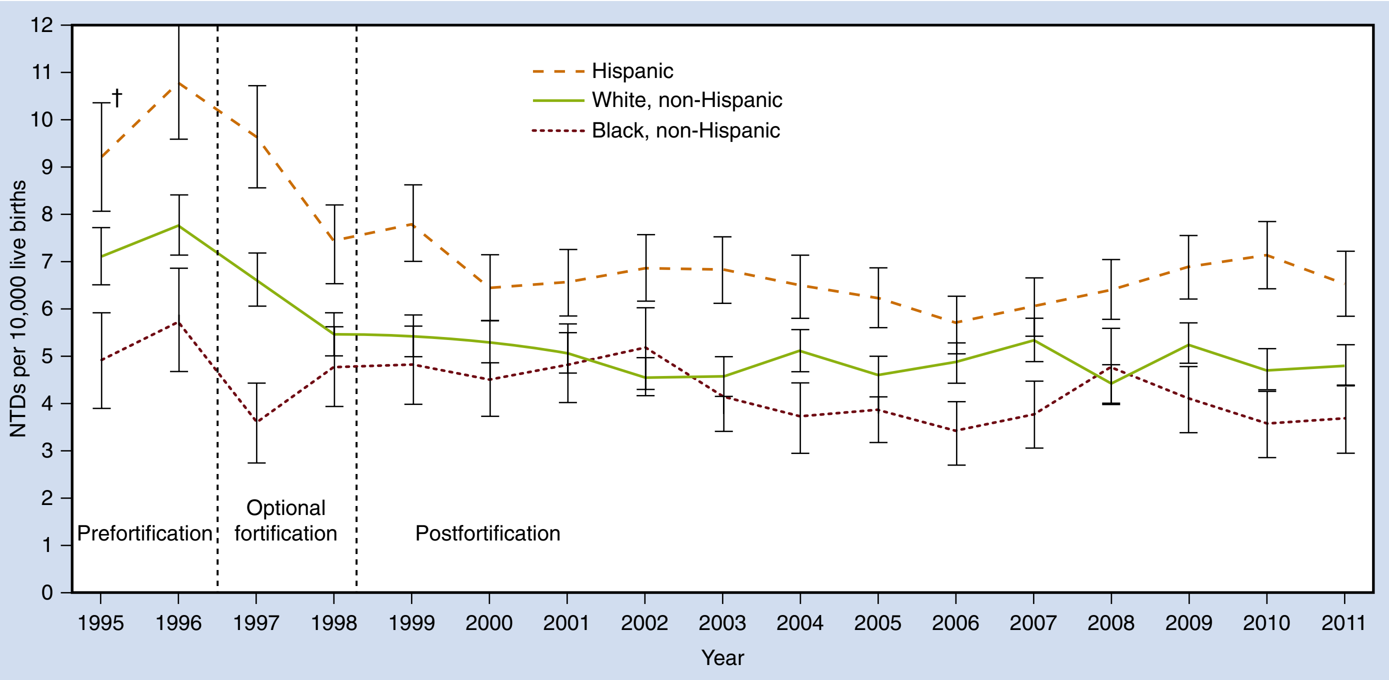
Genetic Factors

There is an increased incidence of NTDs in certain populations even after other factors such as environment are considered, suggesting an underlying genetic cause. For example, the Hispanic population in the United States has a significantly higher prevalence rate for NTDs compared to the White and non-Hispanic populations (Fig. 49.4). These differences persist even with migration to other geographic locations.[45] Next, the risk of bearing a child with a NTD increases in mothers who have previously given birth to a child with a NTD. Liu et al. studied the recurrence rate for NTD in a population in the northern region of China that was known to have a high prevalence rate of NTD. The recurrence rate was about five times greater than the overall prevalence.[69] When evaluating twin births, the concordance for NTDs is higher in same-sex twins compared to opposite-sex twins. Concordance rates between monozygotic twins ranging from 3.7% to 18% have been reported.[28] Lastly, approximately 10% of NTDs occur as part of a genetic syndrome, such as Waardenburg syndrome, Meckel-Gruber syndrome, and Joubert syndrome, or as part of a chromosomal anomaly, such as trisomies 13, 18, and 21.[57] The 22q11.2 deletion syndrome also has a rare association with NTDs. A study of two siblings with 22q11 microdeletion syndrome and MMC revealed an atypical deletion on 22q11.21, providing support for a link with NTDs.[64] However, because the mother also carried this gene deletion and had no clinical signs or symptoms, it is felt there is likely nonpenetrance for clinical expressivity, and other factors play a role in the development of a NTD.[64] Over 60 different genetic mouse mutants have been identified with NTDs. These gene mutations affect multiple different products, including signal transducers, folate binding proteins, tumor suppressor gene products, cytoskeletal components, DNA methyltransferases, nuclear and cell membrane receptors, chromosomal proteins, gap junction proteins, cell surface receptors, and actin regulators and binding proteins.[28] To date, it is unclear if any of these are causative in human NTDs. Research is continuing to investigate potential genetic causes for disruption of folate metabolism or the metabolic pathways that require folate.[28]

Environmental Factors

Folic Acid Deficiency

Folate (vitamin B9) is an essential micronutrient found naturally in a variety of foods, including legumes, eggs, fruits, dark green leafy vegetables, beef, and nuts. It plays an essential role in DNA synthesis, red blood cell formation, new cell formation and differentiation, and metabolic pathways, including homocysteine



• **Fig. 49.4** Prevalence of neural tube defects *(NTDs)* (anencephaly and spina bifida) before and after mandatory folic acid fortification, by maternal race/ethnicity: 19 population-based birth defects surveillance programs, United States, 1995 to 2011. A decline in NTDs was observed for all three of the racial/ethnic groups examined between the prefortification and postfortification periods. Contributing programs are based in Arkansas, Arizona, California, Colorado, Georgia, Illinois, Iowa, Kentucky, Maryland, New Jersey, New York, North Carolina, Oklahoma, Puerto Rico, South Carolina, Texas, Utah, West Virginia, and Wisconsin. †95% confidence interval. (From Williams J, Mai CT, Mulinare J, et al. Updated estimates of neural tube defects prevented by mandatory folic acid fortification—United States, 1995–2011. *MMWR.* 2015;64:1-5.)

regulation, methylation, and nucleic acid biosynthesis.[57] It has a critical role in the formation of the fetal neural system during the first trimester. The association between maternal nutritional deficiency and NTDs was first reported during the Dutch famine from 1944 to 1946 when there was a significant increase in infants delivered with NTDs.[105] This led to further research resulting in the discovery that folic acid (synthetic form of folate) supplementation reduced the prevalence of anencephaly in Ireland and England.[36] Subsequent studies have demonstrated the striking decrease in the prevalence of children born with NTDs following the mandatory supplementation of folic acid in foods in the United States (see Fig. 49.4). Additional studies have evaluated the impact of folic acid supplementation on other risk factors for NTD. The increased risk associated with maternal obesity and pregestational diabetes was reduced with folic acid supplementation of 400 μg.[83,84]

Supplemental folic acid also reduced the risk for NTD in mothers who were exposed to valproic acid during pregnancy.[57] A daily dose of 400 μg of folic acid is recommended for all females of child-bearing age. For females who have had a child with spina bifida, 4 mg of folic acid daily is recommended to prevent future pregnancies complicated by a NTD. Folic acid is a mandated supplement in cereals in the United States and many countries. There is no recommended amount for folic acid supplementation of food, and amounts vary from 0.3 to 5 mg/kg across the world. In the United States, enriched cereals provide about 100 μg of folic acid a day; however, dietary preferences of low-carbohydrate intake or gluten-free diets place potential mothers at risk for low folate levels and bearing children with a NTD.

Maternal Obesity and Maternal Diabetes

Maternal obesity, defined as a body mass index (BMI) of greater than or equal to 30 kg/m^2, is a known risk factor for NTD, although it accounts for only 9.89% of attributable risk in patients with spina bifida in a study by Agopian et al.[2] Hendricks et al.[47] studied the effect of hyperinsulinemia and obesity on the risk for NTD among Hispanic females and questioned if obesity and maternal diabetes were independent risk factors or if hyperinsulinemia associated with obesity and diabetes was the underlying risk factor. Maternal hyperglycemia is known to cause cell death in the neuroepithelium, and maternal hypoglycemia is hypothesized to deprive the embryos of the energy needed for development of the neural tube. Hendricks's study revealed that both hyperinsulinemia and maternal obesity were related to an increased risk for NTD. When adjusted for obesity, the risk for NTD associated with hyperinsulinemia was only slightly reduced. When adjusted for hyperinsulinemia, the risk for NTD associated with obesity was slightly more reduced but still indicated a modest risk with an odds ratio of 1.45.

Prenatal Screening, Diagnosis, and Interventions

Screening

The gold standard for screening and diagnosis continues to be fetal ultrasound. A fetal anomaly scan is typically performed between 18 and 22 weeks of gestation, with close attention paid to the spine and cranium. With the advances in high-resolution and three-dimensional ultrasound, the level of the spinal defect, size of the defect, and level of the conus medullaris can be determined.[19] Specific signs associated with an NTD are listed in Table 49.1. Fetal magnetic resonance imaging (MRI) is being used to further delineate the defect when fetal surgery is being considered. Serum alpha-fetoprotein (AFP) screening is completed between 15 and 22 weeks of gestation. AFP is a protein produced by the fetal liver. Elevations of AFP in maternal blood can be an indication of an open NTD. The detection rate is between 65% and 80%.[19] Maternal AFP levels will vary throughout pregnancy, and an elevated AFP level may indicate an incorrect gestational age. If the maternal serum AFP is elevated, an ultrasound should be completed for further evaluation. With the advances in ultrasound technology, checking amniotic fluid AFP levels is no longer recommended. Early detection allows the medical team to provide education to the parents regarding NTDs, consider the option for fetal surgery, and plan for delivery. A cesarean section for delivery had been considered best practice to prevent further injury to the exposed neural tissues. Older studies had reported an improvement in motor function when babies were delivered by cesarean section and prior to the onset of labor. However, more recent studies report no benefit regarding motor function from cesarean section or absence of labor.[44]

Prenatal Interventions

The surgical approach to MMC was revolutionized with the advent of fetal surgery in the late 1990s. It was hypothesized that intrauterine surgery performed early could prevent some of the damage associated with open flow of CSF from the fetus, exposure of the fetal spinal cord to the amniotic fluid, and traumatic

TABLE 49.1 Ultrasound Findings Associated With Neural Tube Defects

Ultrasound Sign/Finding	Description	Significance
Ventriculomegaly	Enlargement of the ventricles	May indicate hydrocephalus
Dangling choroid sign	In severe ventriculomegaly, the choroid plexus will appear to hang down or dangle in the dilated ventricle	May indicate hydrocephalus
Lemon sign	Concave shape of the frontal bone giving the appearance of a lemon	Associated with Chiari II malformation
Banana sign	Cerebellum appears banana shaped	Indicates caudal displacement of the cerebellum as seen in Chiari II malformation
Saclike protrusions	Fluid filled sacs projecting from the spine	Open neural tube defect, including myelomeningocele or meningocele

injury to the nervous system in utero. The Management of Myelomeningocele Study (MOMS) was commenced in 2003 to examine the outcomes of prenatal repair vs. postnatal repair.[1] In this prospective, randomized controlled study, patients identified for prenatal closure had surgery performed prior to 26 weeks of gestation. Strict inclusion/exclusion criteria included an uncomplicated singleton pregnancy, fetus with an isolated MMC, a normal karyotype, maternal BMI below 35 kg/m^2, and no additional major structural anomalies. The MOMS trial was stopped early for efficacy when initial results demonstrated that prenatal repair of the spinal cord defect decreased hindbrain herniation, decreased the need for shunt placement, and improved neurologic function. The frequency and severity of an Arnold-Chiari II malformation was significantly less in the prenatal group (64% present) compared to the postnatal group (96%). The rate of shunt placement in the prenatal surgical group was 40% compared to 82% in the postnatal group. The MOMS trial showed no difference in infant mortality rates between prenatal and postnatal groups at 30 months of age. Children in the prenatal surgery group were more likely to have a level of function greater than or equal to two levels than the expected anatomic level of defect compared to the postnatal group. Their motor function as measured on the Bayley and Peabody scales was higher than the postnatal group despite having a higher level of lesion.[1] Children in the prenatal surgery group were more likely to walk independently compared to the postnatal group.[1,38] Noted pregnancy complications were more common and included premature delivery with an average gestational age of 34.1 weeks at the time of delivery, oligohydramnios, chorioamniotic separation, placental abruption, and spontaneous membrane rupture. Maternal complications occurred in about one-third of mothers with dehiscence or thinning of the hysterotomy site. Because of known risk of rupture with future pregnancies, it was recommended that all subsequent pregnancies be delivered by cesarean section prior to the onset of labor.[1] The surgical approach in the MOMS trial was an open hysterotomy, and subsequent interventions are assessing the safety and efficacy of a fetoscopic approach as well as a hybrid (open abdominal incision with fetoscopic intervention through the uterine wall) approach. There is not a clear standard of care yet, although the hybrid approach is promising with fewer maternal complications and similar outcomes in the fetus.[118]

The MOMS2 trial[50] evaluated the participants, now ages 5 to 10 years old, in the MOMS study. This study evaluated the motor function, mobility, and ability to perform activities of daily living of the prenatal surgical group and the postnatal surgery group. The prenatal group continued to demonstrate better mobility, gait mechanics, and advanced motor skills, such as stair climbing, than the postnatal group. More children ambulated independently in the prenatal group and required less bracing or assistive device for ambulation. There were fewer knee flexion contractures in the prenatal group, although the presence of ankle plantarflexion contractures was similar. With self-care skills, the prenatal group performed better than the postnatal group in use of a fork, brushing teeth, washing and drying hands, doffing pants and socks, and managing zippers. They were noted to have better dexterity and speed compared to the postnatal group. Other outcomes of the MOMS have reported the need for fewer shunt placements and revisions in the prenatal group up through school age.[118] By school age, 49% of the prenatal group required placement of a shunt compared to 85% of the postnatal group. Shunt revision was required in 23% of the prenatal group compared the 60% in the postnatal group. Urologic function, which did not demonstrate a significant difference in need for clean intermittent catheterization (CIC) at 30 months, demonstrated improvement in the prenatal group by school age. By school age, 61.5% of the prenatal group required CIC compared to 87.2% in the postnatal group. Additionally, 24% of the prenatal group were now able to void volitionally compared to 4.2% of the postnatal group.[20] Interestingly, the rate of augmentation cystoplasty, vesicostomy, and urethral dilation did not differ between the two groups. A final notable outcome following prenatal surgical repair is an increase in the need for release of a tethered cord. By school age, 27% of the prenatal group needed a release compared to 15% of the postnatal group.[118] Despite the releases, the prenatal group maintained better motoric function.

Current research in animal models for fetal MMC repair is investigating the use of transamniotic stem cell therapy without surgery, use of placental mesenchymal stromal cells applied at the time of fetal surgery, use of a cryopreserved human umbilic patch for fetal repair, use of an autologous fetal skin graft in utero, and application of basic fibroblast growth factors on a gelatin scaffold to close the spinal defect in utero. Preliminary outcomes have been favorable.[118]

Neonatal Interventions

With the available screening and early diagnosis of MMC, the multidisciplinary team should be well prepared for the delivery and initial care of the infant. If the child was not a candidate for fetal surgery and is delivered with an open NTD, then early management includes preventing further injury to the neural tissue, preventing infection, preventing complications, and optimizing function. Currently, fetal surgery occurs in less than half of the pregnancies with a diagnosed open NTD.[19] Postnatal surgery should occur within 24 to 48 hours of delivery to decrease the risk of infection and control CSF leakage.[78,91] Most commonly, the MMC presents in the lumbosacral region but can occur at all levels of the spine.[78] If a severe kyphosis is noted, a kyphectomy may need to be performed to allow for repositioning of the neural tissue and skin closure in thoracic level lesions.

Only 10% of infants present with clinically apparent hydrocephalus at birth, but within the first week of birth 85% of infants will demonstrate hydrocephalus.[78] Increasing head circumference and bulging, tense anterior and posterior fontanelles are hallmarks for hydrocephalus. It has been reported that up to 80% of neonates required placement of a ventriculoperitoneal (VP) shunt.[78] Hydrocephalus is directly related to an Arnold-Chiari II malformation, and the infant should be monitored closely for stridor, which can indicate compression of the brainstem. A symptomatic Arnold-Chiari malformation is the leading cause of death in infants with MMC.

In addition to neurologic issues, the newborn should have a full evaluation of the limbs. A clubfoot deformity is most common, though other joint contractures involving the hips and knees may be present. Physical therapy should be involved early to work on range of motion (ROM) and parental education for continued ROM exercises and positioning. Orthoses can be fabricated to assist with maintaining and improving ROM. Orthopedic surgery may be indicated for severe deformities, but typically serial casting is the first line for management.

Early management of bowel and bladder commences with the neurologic examination. Rectal tone should be noted on examination, including if the rectum is patulous. Baseline studies of bladder function, including a renal/bladder ultrasound, urodynamic testing, and serum creatinine, should occur within 3 months of birth.[34] Based on the results of that testing, CIC may be initiated. Indications would include hydronephrosis, vesicoureteral reflux, and inability to void. Between 4% and 9% of infants with MMC will have a high-grade hydronephrosis on ultrasound within the first year of life.[13] For the bowels, the infant should be monitored for constipation. Further discussion regarding bowel management can be found later in this chapter.

Research has demonstrated a high risk for developing a latex allergy in infants with MMC exposed to latex. The prevalence of latex sensitization has been reported to be as high as 72% and varies widely across geographic locations, with the highest prevalence in Europe and the United States.[19] Because of this sensitization, individuals with MMC are at higher risk for anaphylactic reactions to latex. Parental education to avoid latex-containing products is essential. In addition, medical products, including catheters, should be latex free.

Rehabilitation within the neonatal period should include a thorough evaluation of neural function, including a careful motor assessment to determine a neurologic level of function. This can provide some insight into future ambulatory capability. The assessment should also include a review of all comorbidities and their potential impact on function. Optimally, a multidisciplinary team, including neurosurgery, orthopedic surgery, urology, physical medicine, physical and occupational therapy, speech, and nursing, evaluates the infant within the first days of life to establish a treatment plan, obtain needed orthoses or adaptive equipment, and provide family education regarding function, prognosis, preventative care, bowel/bladder management, and resources, including a referral to early intervention services.

Management From Childhood to Adulthood

Comorbidities and Management in Childhood and Adolescence

It is helpful to understand the relationship between the different spinal dysraphisms and their clinical presentations, including associated complications. Table 49.2 provides an overview of the different spinal dysraphisms with some associated comorbidities.

Neurologic

Hydrocephalus

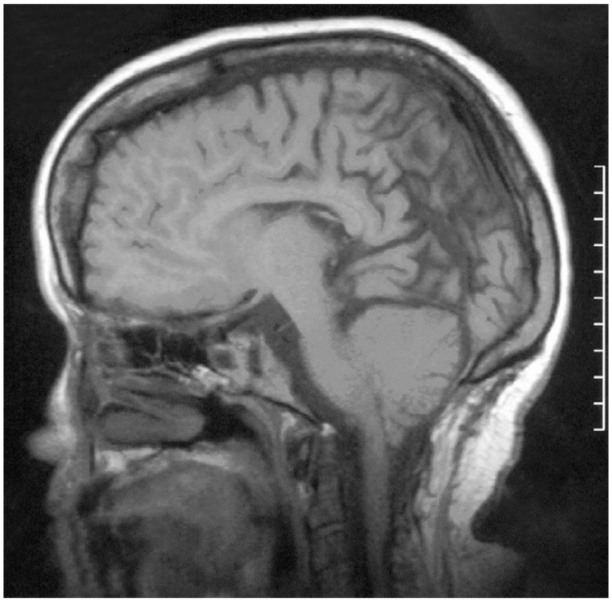
Hydrocephalus is a very common comorbidity in MMC. Historically, 75% to 80% of patients with MMC required shunt placement, but recent advances in intrauterine closure have reduced this burden. When the child has obvious hydrocephalus, some pediatric neurosurgeons prefer to place a shunt concomitantly with primary closure of the MMC.[17,59] There has been a recent trend of temporizing initial treatment of hydrocephalus in less severe cases. The use of external ventricular drains and subgaleal shunts can successfully reduce the need for a shunt.[17,59] Another new alternative strategy to shunt placement is endoscopic third ventriculostomy with choroid plexus cauterization (ETV-CPC). Typically, 90% of the choroid plexus is cauterized. Data from the North American Hydrocephalus Research Network (HCRN) have shown an overall success rate of 45% to 50% of ETV-CPC. Ventricles tend to remain larger following ETV-CPC compared to shunt placement.[17,59]

The focus on brainstem function is critical and a fundamental concept in the treatment of hydrocephalus. An Arnold-Chiari II malformation is easily identified on brain MRI (Fig. 49.5) and is characterized as a caudal displacement of the cerebellar tissue along with the brainstem and fourth ventricle into the spinal canal. Brainstem dysfunction due to a symptomatic Chiari II malformation frequently manifests as poor secretion control and inspiratory stridor. Inspiratory stridor is caused by relaxation of the vocal cords. This sounds like a high-pitched squeak and is a sign of a critically threatened airway. Other signs and symptoms include obstructive or central apnea, bradycardia, dysphagia, cyanosis, nystagmus, vocal cord paralysis, hypotonia, opisthotonus, new or worsening spasticity, upper limb weakness, and torticollis. The child must be monitored closely by medical professionals and treatment promptly initiated. Hydrocephalus due to a shunt failure may be the cause of a symptomatic Chiari malformation and should be investigated first, and posterior fossa decompression should be performed early if needed.[17,59]

Management of hydrocephalus and possible shunt failure as the child grows into childhood and adolescence requires collaboration between the child, caregivers, and medical professionals.

TABLE 49.2 Comorbidities and Clinical Presentation of the Various Spinal Dysraphisms

Spinal Dysraphism Type	Size	Hydrocephalus	Chiari II Malformation	Vertebral Anomalies	Limb Atrophy, Asymmetry, Foot Deformities	At Risk for Tethered Cord Syndrome
Myelocele, myelomeningocele, meningocele, syringocele	Large	+	+	+	+	+
Lipomyelomeningocele, lipomyeloschisis, neurenteric cysts	Large	−	−	+	+	+
Dermal sinus tract	Small	+	−	−	+	+
Split cord malformation, intradural lipomas, caudal regression syndrome, filum terminale fibrolipomas, hamartomas	Small	−	−	+	+	+
Possible delay/failure of fusion of posterior neural arches	Small	−	−	−	−	−



• **Fig. 49.5** Chiari II malformation.

There is no single finding to suggest shunt failure; thus a high suspicion of shunt failure is critical if the child has any change in behavior or mental status. Common signs and symptoms of shunt failure are headache, nausea, vomiting, neck pain, and fatigue. Infants may appear more fussy, sleepy, and unable to console. However, many children will develop symptoms that are unique to them. A unique symptom of shunt failure related to MMC is low back and hip pain. This can mimic tethered cord syndrome.[17,59] Box 49.2 provides possible symptoms of shunt malfunction in each age group.

Tethered Cord Syndrome

Tethered spinal cord syndrome occurs in approximately 40% of patients with open MMC.[5,17,59] It occurs when the spinal cord becomes tethered or fixed at one point causing traction on the spinal cord. It can be seen following both prenatal and postnatal MMC repair as well as in some of the closed spinal dysraphisms, such as lumbosacral lipomatous malformation.[19] When there is concern for a tethered cord, neurosurgical input is required in developing a plan for treatment. Signs and symptoms of tethered cord syndrome include upper motor neuron signs not previously present, such as increased muscle tone or spasticity, brisk reflexes, ankle clonus, weakness, worsening gait, scoliosis, pain, and/or change in bowel or bladder function. Any change in neurologic function requires prompt evaluation and diagnosis to prevent long-term complications.[5,59] Long-term prognosis for motor function and mobility is poor with multiple surgical detethering or if the first surgery is performed at a younger age.[19] Equipment needs for mobility must be assessed after recovery from surgery to determine if additional support is needed. For example, a previously ambulatory child with bracing and forearm crutches may now need a manual wheelchair.

Hydromyelia and Syringomyelia

Hydromyelia (Fig. 49.6) is a fluid-filled cyst occurring in the fourth ventricle and often seen with Chiari II malformations.

• BOX 49.2 Possible Symptoms of a Shunt Malfunction

Infants

- Bulging fontanelle
- Vomiting
- Irritability
- Change in appetite
- Lethargy
- Sunsetting eyes (cranial nerve VI palsy with abduction paralysis)
- Seizures
- Vocal cord paralysis with stridor
- Swelling or redness along shunt track

Toddlers

- Vomiting
- Lethargy
- Irritability
- Seizures
- Headaches
- Swelling along shunt tract
- Redness along shunt tract

School-Aged Children

- Headaches
- Vomiting
- Lethargy
- Seizures
- Irritability
- Swelling along shunt track
- Redness along shunt tract
- Decreased school performance

Adults

- Headaches
- Vomiting
- Lethargy
- Seizures
- Redness or swelling along shunt tract

• **Fig. 49.6** Hydromyelia.

It may be initially asymptomatic, but hydrocephalus or shunt failure may cause progression of the cyst in the central spinal canal and symptoms. Symptoms may include neuropathic pain, numbness, weakness in the upper and lower limbs, scoliosis, spasticity, and headaches. Syringomyelia has a very similar clinical presentation, but the cyst occurs next to the central canal. It can also be potentiated by shunt failure. Symptoms of syringomyelia include neck pain, thoracic back pain, or decline in hand function. Sensory changes include loss of pain, temperature, and proprioception. Children may develop painful dysesthesias. If a syrinx, whether hydromyelia or syringomyelia, is not addressed, atrophy of the hands can take place over time. This can lead to dysmorphic anatomy of the hands secondary to neuropathic arthropathy. Treatment should first address the hydrocephalus whether by shunt placement or revision of a failed shunt. Therapy to work on ROM, strengthening, and spasticity management is helpful.[37,53,101]

Musculoskeletal and Orthopedic

Clubfoot

Foot abnormalities may be as high as 80% to 95% in patients with MMC and can present as a clubfoot, valgus, equinus, calcaneus/calcaneovalgus, and/or talus deformity (Fig. 49.7). The Ponseti method, soft tissue release, osteotomies, and hemiepiphysiodesis are different surgical methods to correct these abnormalities. Complications include wound infection, pressure sores, osteomyelitis, delayed or nonunion of the bones, loss of correction, or avascular necrosis. Complications vary based on the foot abnormality and technique utilized.[26,100,107] Congenital talipes equinovarus (clubfoot) is primarily diagnosed in children with lumbar-level lesions. One study of 37 children who underwent surgical correction of clubfoot demonstrated recurrence of clubfoot in 29 feet by age 2 years. Eighteen of those patients required revision/repeat surgery by age 10 years. Bracing is used in the vast majority of children with clubfoot.[26,100,107] The Ponseti method is often a first-line treatment for congenital clubfoot and should be initiated within the first month of life. The Ponseti method consists of gentle manipulation of the foot and ankle, above the knee casts of the foot/ankle (typically changed weekly) to slowly correct anatomic position over time, percutaneous Achilles tenotomy (PAT), and corrective orthoses.[88] The timing of the PAT is important and is recommended when the foot can be abducted 60 degrees and when there is less than 15 to 20 degrees of dorsiflexion. Following recovery from the PAT, a foot abduction orthosis is recommended to prevent recurrence of the deformity.[88] The goal of stretching, casting, surgery, and/or orthotic use is a plantigrade foot. Given underlying weakness, functional bracing for mobility may be a solid ankle-foot orthosis, knee-ankle-foot orthosis, or hip-knee-ankle-foot orthosis.

• **Fig. 49.7** Clubfoot. A child with a clubfoot, showing a flexed ankle, turned heel, and adducted forefoot. (From Bowden VR, Dickey SB, Greenberg SC. *Children and Their Families: The Continuum of Care*. Saunders; 1998.)

Contractures

Contractures of the lower limbs caused by muscular weakness or imbalance are common complications and can impact ambulation and positioning. Surgical correction of contractures may be helpful if interfering with ambulation, motor development, or bracing. Surgery is typically delayed until the patient is older given the high recurrence rate of contracture in a growing child. Gait analysis is recommended prior to surgery to assess underlying motor function. It is also helpful in determining if bracing is or could be beneficial. For rotational lower limb deformities, twister cables may be considered to assist with ambulation and can be used if a patient is too young for surgical correction. Unfortunately, twister cables are bulky and often difficult to incorporate into a daily routine.[26,101] In those patients with significant external tibial torsion, derotational osteotomies can be considered.[26]

Hips

Muscle paralysis and/or muscle imbalance can result in hip deformities, such as subluxation, dislocation, or hip flexion contracture. Hip flexion contractures contribute to the development of excessive lumbar lordosis, and a hip flexor tenotomy may be beneficial in children who are nonambulatory to improve sitting balance and upright positioning. Hip subluxation and dislocation are common in patients with MMC, and congenital hip dislocation may be present at birth. In the past, surgical intervention to address hip pathology was common. More recently, hip dislocation was not shown to correlate with ambulatory function, and surgical intervention often resulted in a decline in ambulatory function. Therefore surgery to address hip dislocation based on ambulatory needs alone is not recommended. Studies have shown that contracture around the hips has greater impact on function and gait. Current guidelines do not recommend routine radiologic screening of hips except perhaps in the individuals with a sacral-level lesion. Hip subluxation/dislocation can be painful for those individuals who retain sensation in the hip region and may be an indication for medical management and/or surgical intervention. Surgical intervention and other management of hip subluxation/dislocation should be individualized based on the clinical presentation and symptoms.[11,99,101]

Spine

Patients with MMC can develop thoracic kyphosis, lumbar lordosis, or scoliosis. These can affect cosmetic appearance, ambulation, transfers, wheelchair mobility, and quality of life. It can lead to other consequences such as back pain, difficulty with catheterizing or bowel program, and less social and personal acceptance. Spinal abnormalities may be congenital, neuromuscular, or a combination of both.[54,99,101] The presence of scoliosis is generally related to the level of neurologic impairment. When scoliosis presents at a young age, breathing and pulmonary function should be closely monitored.[99,101] Bracing has not been shown to be effective in halting or slowing the progression of neuromuscular scoliosis. If bracing or casting is tried, special considerations must be made. Children with insensate skin are at a higher risk for skin breakdown while in the brace.[101] Once a spinal abnormality is identified, radiographic monitoring should be performed every 1 to 2 years.[101] There is no current consensus regarding surgical intervention for scoliosis in MMC. Treatment decisions, including surgical options, should be determined on an individual case-by-case basis. Surgical management of scoliosis in MMC carries significant risks that must be identified. Infection, worsening of mobility, pseudoarthrosis, and respiratory complications are possible risks. Patients with already lowered or compromised pulmonary function are at risk for postoperative respiratory complications.[99,101]

Mobility

"Will my child walk?" This question is of paramount importance to families and one of the first questions they may ask. For the treating team, this can have many different meanings. Ambulation can be therapeutic or functional and may include the use of bracing and assistive devices (Box 49.3). Helping a family understand that mobility may include walking and/or wheelchair mobility requires time and patience as the child grows and develops. It is also important to remember that mobility may change with growth, obesity, and complications such as shunt failure or tethered cord syndrome. When developing expectations regarding motoric control and functional abilities, the level of the lesion assists with prognostication. However, many children will have functional levels that are better than expected given the anatomic level or may be asymmetric in motor and sensory function. The presence of spasticity or flaccidity will also affect function. Table 49.3 provides expectations for motor function, mobility function in childhood through adulthood, commonly used equipment, and recommended orthoses for the different anatomic levels of lesion.[8,16]

Orthoses to improve anatomic position, such as the foot abduction orthosis, have been discussed previously in the management of a clubfoot deformity. For functional mobility, including standing, transfers, and ambulation, there are many options available. (See Chapter 13 for a review of lower limb orthoses and their applications.) However, two orthoses are primarily used by children with MMC. The first, a parapodium (Fig. 49.8), allows a child even with a thoracic-level lesion the opportunity to be mobile in an upright position. By rotating the body, the child can advance the parapodium and learn to weight shift. The parapodium also allows a child to use the arms freely in a standing position.[23] The other is a reciprocating gait orthosis (RGO), which adds a cable system to the hip-knee-ankle-foot orthosis. The child must have active hip flexion to activate swing phase and advance one foot at a time. It is more energy and time consuming than a swing-through gait pattern, but studies have indicated that some children prefer this orthosis. Neither the parapodium nor the RGO tend to be used into adolescence or adulthood, but the early mobilization of younger children to be able to interact with their peers and environment has been shown to improve cognitive and psychosocial skills. In general, most individuals with a sacral level of lesion are community ambulators and most with a thoracic level of lesion are nonambulators. The most important predictors for ambulation are quadriceps and iliopsoas strength and sitting balance. The functional mobility scale is a nice tool to track mobility over time (Fig. 49.9). A decline in rating should be investigated for possible treatable complications, including shunt failure, hydrocephalus, tethered cord syndrome, contractures, and obesity. It may also be used as an outcome measure following surgical procedures.

• BOX 49.3 Hoffer Ambulation Criteria

Hoffer Classification of Ambulation and Criteria

- **I. Community ambulators:** Patients walk indoors and outdoors for most activities; may need crutches, braces, or both. Wheelchair used only for long trips out of the community.
- **II. Household ambulators:** Patients walk only indoors and with orthoses. Able to get in and out of chair and bed with little, if any, assistance. May use wheelchair for some indoor activities at home and school. Wheelchair is used for all activities in the community.
- **III. Nonfunctional ambulators:** Patients walk during therapy session at home, in school, or in hospital. Wheelchair used for all other transportation.
- **IV. Nonambulators:** Patients are mobile only via a wheelchair but usually can transfer from chair to bed.

Gastrointestinal/Urinary

Catheterization/Bladder Program

The most common cause of neurogenic bladder in children is lumbosacral MMC, with up to 98% of children having signs and symptoms of neurogenic bladder. Neurogenic bladder can lead to urinary incontinence, urinary tract infections, and vesicoureteral reflux. Long-term complications can lead to renal scarring and eventually renal failure if not properly managed.[94,101,108] The main goals of treating neurogenic bladder are to preserve renal function and improve bladder function and continence. Proactive treatment is preferred in managing neurogenic bladder to prevent complications. This is typically done with a CIC program. Pharmacologic management is also often used.[40,94,108]

Children with neurogenic bladder can have reduced bladder sensation or no sensation at all. Therefore they are not aware of bladder filling or may have no desire to void. Reflexia of the detrusor muscle varies between 13% and 49.5% and hyperreflexia between 25% and 76%. One study of 112 children who underwent urodynamic studies showed only 4 children with normal bladder function. Therefore it is recommended that all children with MMC undergo baseline urodynamic evaluation. It is also recommended to perform renal ultrasound to detect hydronephrosis or abnormal anatomy.[94,101,108]

Anticholinergic medications reduce intravesical pressure. Oxybutynin is not approved in children younger than 5 years but is commonly used. A multicenter retrospective study showed reduction of mean end filling detrusor pressure. There were no reported serious side effects, but common side effects of anticholinergic medications, such as constipation, facial flushing, and dry mouth, should be noted.[40,94,101] B3 agonists, such as mirabegron, help to relax the detrusor muscle. Several studies in pediatric populations have demonstrated positive effects of this medication, including voided volume, urinary continence, and increased compliance with a bladder program. There has been no evidence that alpha blockers aid in the management of neurogenic bladder in children with MMC.[94] Botulinum toxin type A to the detrusor muscle can be considered. Studies demonstrate beneficial effects in clinical and urodynamic variables. Botulinum toxin increases continence, mean bladder capacity, and mean maximum cystometric capacity. As with other injectables, the benefit of botulinum toxin is transient, and most patients will require repeat treatments, but no sooner than 90 days after their last injection. Side effects of botulinum toxin are low but include cases of mild hematuria and urinary tract infection.[93,101] Caregivers for young children with MMC should be trained on how to properly perform CIC by a trained professional. As they mature, children can learn how to perform self-catheterization to improve independence. Timing of

TABLE 49.3 Grouping and Prognostication of Motor Function by Level of Lesion

Level of Lesion	Expected Muscle Function	Functional Mobility in Childhood	Functional Mobility in Adolescence and/or Adulthood	Commonly Used Equipment	Orthoses Used	FMS Classification	Functional Pattern
Lower thoracic (T11–T12) **Functional hallmark**	May have some abdominal, paraspinal, and quadratus lumborum function **No lower limb motor function, lack iliopsoas function**	Therapeutic standing[a] Nonfunctional ambulation[b] Complete wheelchair reliance for mobility	No ambulation Complete wheelchair reliance for mobility	Standing frame[a] Parapodium[b] Wheelchair	Trunk-hip-knee-ankle-foot orthosis	FMS: 1, 1, 1	**Group 1**[a]
High lumbar (L1–L2) **Functional hallmark**	L1: Iliopsoas group 2 or better L2: Iliopsoas, sartorius, and adductors all group 3 or better May have weak quadriceps Some sensation present below hip joint **Have iliopsoas function**	Wheelchair reliance for distance ambulation Limited household ambulation[c]	Complete wheelchair reliance for distance ambulation May exhibit nonfunctional ambulation	Wheelchair Forearm crutches Walker	Reciprocating gait orthosis Hip-knee-ankle-foot orthosis	FMS: 1, 1, 1	**Group 1**[a]
Midlumbar (L3–L4) **Functional hallmark**	L3: Meet criteria for L2 and quadriceps is group 3 or more L4: Meet criteria for L3 with strong quadriceps and have moderate medial hamstrings **Have strong knee extension** **Lack hip abduction**	Household ambulation, limited community ambulation Partial reliance on wheelchair for distance ambulation	50% wheelchair reliance Usually maintain household ambulation with forearm crutches	Wheelchair Walker Forearm crutches	Knee-ankle-foot orthosis	FMS: 3, 2, 1	**Group 2**[b]
Low lumbar (L5) **Functional hallmark**	L5: Meet criteria for L4 and have strong medial and lateral hamstrings and tibialis anterior and posterior, have moderate hip abduction **Strong knee flexion, strong ankle dorsiflexion and inversion, and great toe extension** **Lack active hip extension and ankle plantar flexion**	Community ambulation Partial reliance on wheelchair for long-distance ambulation	Community ambulation with forearm crutches Partial reliance on wheelchair for long-distance ambulation	Wheelchair, forearm crutches	Ankle-foot orthosis	FMS: 6, 6, 6 But may be FMS: 6, 6, 3, or 4 in adulthood	**Group 2**[b]

TABLE 49.3 Grouping and Prognostication of Motor Function by Level of Lesion—cont'd

Level of Lesion	Expected Muscle Function	Functional Mobility in Childhood	Functional Mobility in Adolescence and/or Adulthood	Commonly Used Equipment	Orthoses Used	FMS Classification	Functional Pattern
Sacral (S1) (S2–S3) **Functional hallmark**	S1: Meet criteria for L5 Have normal ankle dorsiflexion, inversion, eversion, and moderate ankle plantar flexion and hip extension S2–S3: Meet criteria for S1 and have gastrocnemius and soleus **Have active hip extension and ankle plantar flexion and eversion** **May lack foot intrinsic muscle function**	Community ambulation Community ambulation	Community ambulation, partial reliance on crutch or cane in adulthood for long distances Community ambulation, may have reduced endurance and pain as a result of late foot deformities	None None	Supramalleolar foot orthosis Foot orthosis	FMS: 6, 6, 6	**Group 3**[c]

[a]Group 1: Generally nonambulatory/nonfunctional ambulation unless they have pelvic elevator strength (quad lumborum), in which case they may attain household ambulation; completely wheelchair reliant as adults.
[b]Group 2: Community ambulation with braces and assistive device; partial reliance on wheelchair.
[c]Group 3: Community ambulation with foot orthoses and without need for assistive device or wheelchair.
FMS, Functional Mobility Scale.
Data from Apkon SD, Grady R, Hart S, et al. Advances in the care of children with spina bifida. *Adv Pediatr.* 2014;61(1):33-74; Bartonek A, Saraste H, Knutson LM. Comparison of different systems to classify the neurological level of lesion in patients with myelomeningocele. *Dev Med Child Neurol.* 1999;41:796-805; Broughton NS, Menelaus MB, Cole WG, et al. The natural history of hip deformity in myelomeningocele. *J Bone Joint Surg Br.* 1993;75:760-763; Mazur JM, Shurtleff D, Menelaus M, et al. Orthopaedic management of high-level spina bifida. Early walking compared with early use of a wheelchair. *J Bone Joint Surg Am.* 1989;71:56-61; McDonald CM, Jaffe KM, Mosca VS, et al. Ambulatory outcome of children with myelomeningocele: effect of lower extremity strength. *Dev Med Child Neurol.* 1991;33:482-490; Sharrard WJW. The segmental innervation of the lower limb muscles of man. *Ann R Coll Surg Engl.* 1964;35:106-122; Swaroop VT, Dias L. Orthopaedic management of spina bifida. Part II: foot and ankle deformities. *J Child Orthop.* 2011;5:403-414.

catheterization depends on age, bladder capacity, and fluid intake. Males typically have an easier time learning how to perform self-catheterization. Females may need to use assistive equipment such as a mirror to help locate proper anatomy. Hand washing should be performed prior to any catheterization.[94,101]

Bowel Program

Neurogenic bowel affects most individuals with MMC. Abnormal colorectal anatomy, slowed colonic transit, alterations in sensation, and dysfunction of sphincters contribute to bowel dysfunction.[10,58] Lesions above the conus medullaris result in an upper motor neuron (i.e., spastic) bowel with constipation. Lesions below the level of the conus medullaris result in lower motor neuron (i.e., flaccid) bowel with leaking and incontinence.

Neurogenic bowel can result in incontinence and/or constipation. Incontinence has been associated with several health-related quality of life measures, including depression, discrimination by peers, decreased school/work attendance, lower educational attainment, and lower rates of employment. This can also lead to emotional, physical, and psychological distress. Up to 80% of patients with neurogenic bowel from MMC use some type of bowel program. Without proper bowel management, there can be several complications, including urinary incontinence, urinary tract infections, ventriculoperitoneal shunt malfunction, skin breakdown, hemorrhoids, and anal fissures. There is no standard consensus for a bowel program, and individualized plans can be made for each patient.[10,58,101] Diet with increased fiber as well as adequate fluid intake can help to promote stool consistency. These can help to ensure stool is not too hard, resulting in constipation, or too loose, resulting in possible incontinence.[10,58,101]

There are many different options for management of neurogenic bowel in MMC. Antegrade enemas and oral medications are most commonly used. Antegrade enemas are reported to have the highest usage percentage and lead to the highest rate of continence. Cone/balloon enemas had the second highest rate of continence in children. When enemas and oral medications were used in combination, the rate of continence increased.[6,10,58,101]

Certain medications should be avoided in patients with MMC with altered renal function. Sodium phosphate enemas should be avoided because they can lead to hyperphosphatemia, hypocalcemia, and hypokalemia. Magnesium-based enemas should also be avoided due to the risk of hypermagnesemia and hypocalcemia.[6,101]

• **Fig. 49.8** A parapodium can allow children a degree of motion and flexibility, which has been shown to improve cognitive and psychosocial skills. (Courtesy Orthos International B.V., Amersfoort, The Netherlands.)

A high percentage of those with neurogenic bowel are reliant on a caregiver to help administer their bowel program.[58,101] A Peristeen transanal irrigation system can promote independence with a bowel program. A rectal catheter is inserted into the rectum, and a pump system is then activated to first inflate the rectal balloon, then instill water into the rectal vault and colon, followed by deflation of the balloon to allow evacuation of the stool and water. This system can increase independence for those who cannot administer their own enemas independently. Different solutions, including water or saline, can be used. There is no current universally recognized standardized solution type or weight-based dosing.[58]

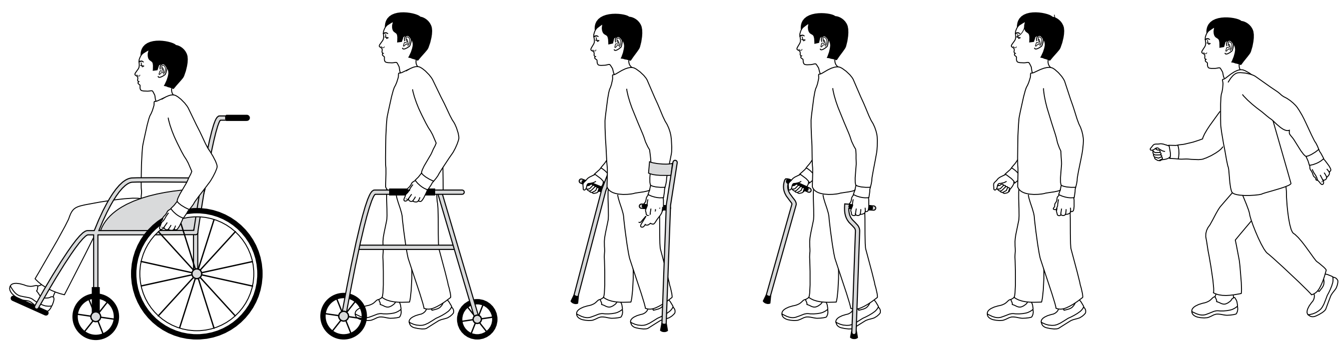
Renal Function

Management of children with neurogenic bladder from MMC should be focused on preservation of renal function, prevention of renal scarring, and preventing progression to chronic renal disease. Early management has been shown to have the best benefits in avoiding long-term complications. Approximately 10% to 30% of children with MMC are born with evidence of renal disease. Accurate measures of renal function are essential. Creatinine, glomerular filtrate rate, and protein excretion are markers to evaluate for renal dysfunction and damage.[56,76]

Surgical Interventions for Bladder and Bowel Care

The creation of a catheterizable channel can help to improve quality of life. These channels are often easier to access, making CIC easier to perform. There are several different surgical techniques currently available. The Mitrofanoff appendicovesicostomy uses the cecal appendix, if still present, to form a channel into the bladder. The

Functional Mobility Scale



Rating 1	Rating 2	Rating 3	Rating 4	Rating 5	Rating 6
Wheelchair reliant for all ambulation	Independent ambulation with walkers or frame	Independent ambulation with bilateral crutches	Independent ambulation with single crutch or bilateral/single hand-held cane	Independent ambulation on level surfaces without assistive device	
May do some supported standing or take steps with walker/hand-held support to facilitate transfers			Holds on to furniture/ walls for support in home	Requires rail for stairs, may need help running	

• **Fig. 49.9** Functional Mobility Scale. Scoring of mobility over three distinct distances representing mobility in the home, school, and community. (Data from Battibugli S, Gryfakis N, Dias L, et al. Functional gait comparison between children with myelomeningocele: shunt versus no shunt. *Dev Med Child Neurol.* 2007;49:764-769; Guidera KJ, Smith S, Raney E, et al. Use of the reciprocating gait orthosis in myelodysplasia. *J Pediatr Orthop.* 1993;13:341-348; Uehling DT, Smith J, Meyer J, et al. Impact of an intermittent catheterization program on children with myelomeningocele. *Pediatrics.* 1985;76:892-895.)

Yang-Monti channel uses the reconfigured ileum. The spiral Monti creates a longer tube that can be used in the case of obese patients.[62,65] There are surgical augmentative procedures to assist with a bowel program, including sacral nerve stimulation, bowel resection, Malone appendicostomy, and diverting colostomy. These have not been studied well in children with MMC.[101,102]

Skin

Skin injuries are very common in the MMC population whether due to pressure, friction, heat, cold, or trauma. Approximately one in five children with MMC will develop a pressure injury. Reduced sensation, impaired mobility, cognitive impairment, inadequate nutrition, and incontinence increase the risk for skin issues and pressure injuries. Pressure injuries occur over bony prominences when the pressure prevents blood flow to the area, thus decreasing oxygen and nutrient supply leading to cell damage and cell death. Those who are nonambulatory typically develop pressure injuries over the sacrum, ischial tuberosities, and greater trochanters. This is complicated by shearing forces related to repositioning and transfers as well as moisture due to incontinence. In a child with a thoracic level of injury, a gibbus deformity may be present, and the skin over this area is also at risk for breakdown. Similarly, a scoliotic curve may increase the risk of skin injury over the apices of the curve or related to the hardware implanted with a spinal fusion. Seating surfaces, including in the wheelchair, bath equipment, toileting equipment, and bed, should be optimized for pressure relief in addition to education to the child and family regarding timing of and performance of pressure relief. In the ambulatory population, skin injuries typically occur on the feet and are likely due to a poorly fitting orthosis. Children and their families should be taught to inspect the skin daily, paying close attention to bony prominences. Pressure injuries are associated with pain, depression, reduced participation, and can lead to sepsis, amputation, and even death. They are often difficult and costly to treat. Studies demonstrate an increased length of stay for a child with MMC admitted to the hospital with a pressure injury.[86,101,104] When a pressure injury is identified, it should be addressed in a multidisciplinary fashion. Identifying the source or cause of the pressure injury and subsequent removal of the cause are essential to ensure proper healing and prevent recurrence. Specialized fluidized beds and rotating beds are recommended to keep the pressure off the injury if located over the back or pelvis. Once a pressure injury occurs, that area will be at higher risk for recurrence even after healing. Some injuries may require debridement, a wound vac, and/or a surgical flap.[86,101,104]

General Medical

Endocrine

Short stature in children with spina bifida has been widely reported and is most commonly due to short growth in the lower limbs. The etiology of short stature is multifactorial and includes hip dysplasia, scoliosis, joint contractures, possible neurotrophic factors, sensory impairment due to injury of the spinal cord, nutrition, and shortened lower limbs.[80] True growth hormone (GH) deficiency is estimated to occur in 30% of children with spina bifida. Hydrocephalus impacts the secretion of pituitary hormones responsible for growth and puberty through abnormal hypothalamic-pituitary function. The exact mechanism is currently unclear. To determine if a child may have GH deficiency, measurements of arm span over time to determine height is recommended. Arm span measurements remove the confounders of scoliosis and short lower limbs. A child with a shorter arm span and decreased rate of change in growth over time compared to peers is likely to have GH deficiency. A referral to a pediatric endocrinologist should be made when this is recognized. Studies have shown that treatment with human GH can result in a significant gain in height. However, the child should be monitored for possible side effects of GH therapy, including new or progressive scoliosis, need for VP shunt revision, tethered cord, and disproportionate growth. Increased growth leads to increased weight, which can cause altered gait mechanics if ambulatory. It can also make transfers more difficult for the patient or caregiver. The patient and the family should have clear goals regarding growth and be made aware of the possible side effects.[8,33,43,70,80] GH deficiency also causes decreases in bone mineral density and muscle mass as well as dyslipidemia.[80]

Central precocious puberty has been reported to occur in females with MMC with a prevalence as high as 50% and in males with MMC with a prevalence of 10% to 30%.[4] It is defined as the onset of pubertal changes in females younger than 8 years old and in males younger than 9 years. These ages of onset may vary based on the child's ethnicity or race. In females, these changes include progressive breast development, maturation of the nipples and areolae, or vaginal secretions. Hair in the pubic or axillary areas may or may not be present. In males, findings may include bilateral testicular enlargement, penile enlargement, and axillary and pubic hair. If left untreated, the growth plates will close sooner than expected resulting in decreased adult height.[4] The psychosocial impact is significant, with an increased risk of depression, substance use, anxiety, poor body image, low self-esteem, and early sexual behavior. Though the cause for precocious puberty is unknown, hydrocephalus, increased perinatal intracranial pressure, and Chiari II malformations have all been associated.[4] A gonadotropin-releasing hormone (GnRH) stimulation test will confirm the diagnosis and treatment with long-acting GnRH analogs is recommended.[4]

Obesity

Children with MMC have a lower baseline metabolic rate and decreased energy expenditure, which can predispose them to obesity. Caloric intake of 80% of the recommended daily allowance has been shown to maintain weight in children with MMC. On average, the child with MMC expends 25% less energy at baseline compared to a child without MMC. The rate of obesity in this population is comparable to that in the general population and has been reported in children as young as 6 years. Obesity has been associated with a decline in function and can lead to the medical problems of metabolic syndrome, including hypertension, cardiac disease, apnea, diabetes, and hyperlipidemia. Interestingly, some studies have shown that increased BMI can have a protective effect on bone and fractures in certain body regions. The reduced risk is in the hip and pelvis, whereas there is an increased risk of fracture in the extremities with increased BMI. Fat may also exert hormones such as insulin, amylin, and leptin through the increased aromatization of testosterone to estrogen, which can have positive effects on bones.[41,42,75] A consistent approach to exercise/activity and nutrition is recommended. There is not yet literature regarding the use of GLP-1 receptor agonists in this population.

Psychological and Social Issues

Intellectual Disability

There have been many studies evaluating the intelligence quotient (IQ) in patients with MMC. They demonstrate that hydrocephalus and shunted hydrocephalus have a negative effect on intelligence in those with MMC. Children with MMC and hydrocephalus have low to normal intelligence compared to children with a closed spinal dysraphism without hydrocephalus who performed similarly to their typically developing peers with a normal IQ. Up to 30% of children with MMC will be diagnosed with an intellectual disability (IQ <70).[66] Repeated shunt revisions and complications from the shunt also correlate with lower IQ scores compared to those who do not have these complications. Despite this impact on intelligence, hydrocephalus must be identified and appropriately treated to avoid other neurologic complications.[3] Children with a higher spinal lesion level tend to have lower IQ scores compared to those with a lower-level lesion. Those with a higher lesion level also have a higher infection rate, which can lead to lower intelligence.[3]

It has been observed that children with MMC have atypical organization of their cerebral cortex. There is a negative correlation between intelligence and increased cortical thickness in children with MMC. Increased cortical thickness was also related to poor fine motor and cognitive functions. Conversely, a low cerebral cortical thickness was associated with a higher level of intelligence.[3]

Neuropsychology

Neuropsychological studies have shown a pattern of strengths and weaknesses involving children with spina bifida. These patterns are associated with an open spinal lesion usually in combination with a Chiari II malformation. Motor, cognitive, academic, and adaptive functions can be affected.[87] Their neurocognitive pattern typically involves strengths in learning skills and performing tasks that rely on associative, rule-based processing (word reading, fact retrieval). Vocabulary, grammar, and word recognition are strengths when it comes to reading and writing.[87] Weaknesses are present when learning and performance involve the construction or integration of information (reading comprehension, math problem solving).[87] Their verbal skills are significantly better than visual perceptual skills, even at a young age. For example, children can learn motor skills through verbal and tactile correction of errors and repetition, which involves the basal ganglia if preserved. However, they have difficulties performing a motor task that requires matching the motor task to visual information involving the cerebellum.[87] They can learn math facts but struggle with complex problems that require multiple steps and algorithms. They can also have problems with estimation.[87]

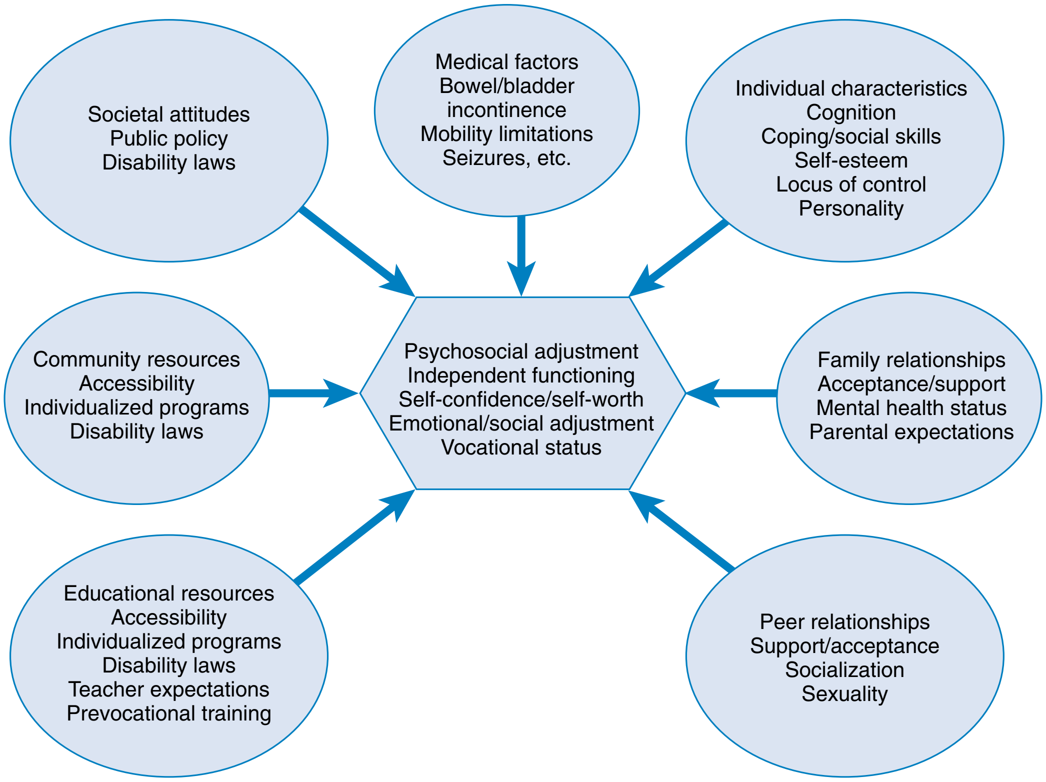
Many children with spina bifida meet the diagnostic criteria for attention-deficit/hyperactivity disorder (ADHD) inattentive type. In contrast to children with developmental forms of ADHD related to self-regulation, the profile of children in the spina bifida population are characterized by underarousal and distractibility. Disruptions in the midbrain and posterior cortex cause difficulties in alerting and orienting to external stimuli. These deficits are noticeable from infancy and involve posterior brain pathways in contrast to the frontal lobe dysfunction seen in developmental ADHD, traumatic brain injury, and the like.[87] In addition to challenges with attention, children with spina bifida demonstrate significant difficulty with auditory and visual memory. Working memory and processing speed are also impaired.[66]

The rehabilitation team should provide parents with educational tools to use in the first year of life to promote parent-child interactions at a developmentally appropriate level. Infant developmental assessment scales can assess motor, cognition, language, and social skill levels. Intervention should be targeted to areas of deficit, and a referral to an early intervention program at 1 to 2 years of age is recommended. Once old enough, a special education preschool can provide additional services to work on cognitive, academic, language, and social skills. If available, a full neuropsychological evaluation at 5 to 6 years of age will identify the child's areas of strengths and weaknesses to better develop a school intervention plan (IEP or 504 plan). It can guide teachers, parents, and therapists in helping the child learn and attain academic and social skills. Attention deficit should be recognized early, and accommodations in the school and home should be provided to assist with keeping on task and learning. Follow-up neuropsychology testing to include literacy and numeracy skills as the child progresses through school is recommended to address new or continued areas of weakness and adjust the educational plan. For those children with intellectual disability, school-based programs to provide additional support for reading and math may be available. If a child is diagnosed with an intellectual disability, this should be documented within the school assessment/record prior to the age of 18 years. This allows for continued support and education as an adult. Screening for anxiety and depression should occur annually, and appropriate referrals for counseling or medication management should be completed. For those children who will need additional help as an adult due to their cognitive impairments, guardianship or power of attorney are options that should be completed by age 18 years.[87]

Behavioral

Children with MMC have very characteristic personality traits and are often described as having a "cocktail" personality. Their verbal skills are often much better than their visuoperceptual skills. However, they have deficits in the ability to process information and make decisions based on that information. Many studies have noted that children and adults with MMC are able to express language correctly as far as syntax and grammar but have difficulty with understanding language context and social cues.[66] Thus they often enter or start conversations easily but may be off topic and lose the thread of the conversation. Holbein et al.[49] studied a group of children ages 8 to 15 years old and noted they had less adaptive social behaviors in peer interactions related to their difficulty with reciprocal verbal exchanges. There were noted deficits in language processing, conversational skills, and social cognition. A child's ability to speak in full sentences and with appropriate phrases often leads others to incorrectly overestimate their cognitive abilities and understanding, especially at a young age.[66] Children with a shunt have more difficulties in this area compared to those without a shunt.

These deficits in executive functions, language comprehension, and social cognition result in children and adults being more withdrawn, less adaptable, and less involved in their community.[66] Children have difficulty with social skills related to social immaturity and passiveness. They have fewer friends and fewer social contacts outside of school. Symptoms of depression and anxiety have been reported, and teens and adults report dissatisfaction with their interpersonal relationships, employment opportunities, and recreation. This correlated directly with their IQ.[48] Individuals with MMC commonly need assistance with managing their finances, educational tasks, and household tasks.[66] When developing a rehabilitation plan, it is important to remember the many factors that affect psychosocial functioning and adjustment (Fig. 49.10). It is



• **Fig. 49.10** Factors influencing psychosocial adjustment and disability in individuals with spina bifida. (Modified from Ayyangar R. Health maintenance and management in childhood disability. *Phys Med Rehabil Clin N Am.* 2002;13:793-821.)

strongly recommended to address these cognitive and language deficits early through neuropsychology testing, speech therapy, and counseling along with involvement of the family and school.

Adult Care

Adults with spina bifida must deal with the normal physiologic issues related to the normal aging process in addition to the associated secondary complications of their disability. This interrelationship has some similarities to that experienced by the population aging with spinal cord injury. It is now believed that these complications previously thought to be static in nature can have an increasing impact on functional ability with age. In treating the adult patient, one must evaluate the usual age-related medical problems as well as the issues unique to this population.[60] While most children with NTDs are followed closely by their pediatric providers and often in multidisciplinary clinics, adults with NTDs often do not have regular medical follow-up that is comprehensive in nature. In comparing outcomes in young adults who had been followed in a multidisciplinary clinic vs. those who lacked coordinated care, it was clear that outcomes were considerably worse in the latter with over 66% of the adults lacking regular medical and specialty care.[63] While transitions of care to adult providers and adult models of care remain difficult, some success with coordinated systems for transition have been described and implemented.[67]

General Medical Considerations

Adults with NTDs have ongoing increased medical needs compared to the general population, including spina bifida–specific care, age-related secondary disabilities, and general adult medical needs.[114] Individuals with NTDs are at higher risk for hypertension, and this may require treatment as early as in the teenage years. They also have significant risks for obesity and sleep apnea.[68] Obese adolescents and adults with NTDs have an increased prevalence of components of the metabolic syndrome. Adults and adolescents with MMC have decreased aerobic fitness and muscular strength, decreased lean mass, and increased fat mass, all of which put them at higher risk of developing metabolic syndrome and cardiovascular diseases.[22,42] Admission rates to acute hospitals for adults diagnosed with disabilities in childhood, including MMC, are nine times higher than that of the general population, and 76% do not have or cannot identify a primary care provider. For adults with MMC, the most common primary diagnosis for hospitalization was urinary tract infection, followed by complications from devices/grafts/implants, and skin wounds. Sepsis accounted for the most deaths. Approximately one-third of hospitalizations were for primary diagnoses of potentially preventable conditions.[30] Multidisciplinary clinics for adults with MMC, similar to those in pediatric institutions, have the potential to reduce these hospitalizations. Unfortunately, there are far fewer clinics dedicated to treating adults with MMC than those that treat only pediatric patients.

Urologic Considerations

Despite the high prevalence of urologic complications in patients with NTDs, the majority do not receive the minimum recommended follow-up. Complications can include chronic incontinence (often contributing to pressure injuries), cystitis, chronic kidney disease, and pyelonephritis.[95] Renal damage remains one of the most common causes of morbidity and mortality among individuals with

NTDs with an incidence that is significantly higher than individuals without NTDs.[115] Adults with neurogenic bladder and NTD are at an increased risk for bladder cancer. Those who have had bladder augmentation with ileal or colonic segments have a 7- to 8-fold risk, and those with gastric augmentation have a 14- to 15-fold risk for the development of bladder cancer compared with the population at large. Individuals who have used a Foley catheter for over 10 years are also at risk of bladder cancer and need to be monitored consistently.[52]

Musculoskeletal Considerations

Data on late musculoskeletal conditions in the MMC population are sparse; therefore much of the treatment recommendations are gleaned from data on the spinal cord injury population. Shoulder pain is very common in all long-term wheelchair users, although it may be less common in patients who began using a wheelchair earlier in life.[96,110] Rotator cuff disorders and bicipital tendonitis are the most common injuries reported in chronic wheelchair users.[39] The Myelo Clinic team must work together from an early age to provide education and training to patients with MMC who are wheelchair users to reduce the risk of shoulder pathology. Charcot neuropathic osteoarthropathy is a destructive joint disorder initiated by trauma to a neuropathic extremity.[46] In patients with MMC, this can occur in hips, knees, and, most commonly, ankles and feet. Charcot joints can lead to progressive loss of function and interfere with positioning and activities of daily living. Bracing to prevent progressive deformity is indicated. The ambulation status of adults with spina bifida has been shown to deteriorate over time. This loss of function is multifactorial, including spasticity, progressive joint contractures, low back pain, medical comorbidities, and patient preference.[29]

Sexuality, Sexual Health, and Fertility

There are few studies available to guide clinicians in caring for and advising patients with MMC on issues of sexuality, sexual health, and fertility. However, it is important that these issues be addressed. Both males and females with MMC are highly likely to have decreased sensation in the perineum, which can impair the ability to reach orgasm.[27] In addition, nonverbal learning disorders and societal attitudes toward individuals with disabilities can affect self-esteem, social interactions, and ultimately psychosexual development.[97] This may be more of an issue for males than females because females with MMC were reported to be more than twice as sexually active as males with MMC.[111] The reported incidence of the ability to achieve erection ranges from 14% in those with higher-level lesions to 85% in those with lower-level lesions, although many of these achieved reflexively.[97] The ability to sustain erection during sexual activity is uncertain, and many patients report dissatisfaction with the degree of rigidity. Successful treatment of erectile dysfunction in MMC with phosphodiesterase inhibitors (such as sildenafil) has been reported.[82] Less commonly, surgical reinnervation of the penis has shown success.[81]

Satisfaction with sexual function among individuals with MMC is unclear. A single, small (53 individuals, both male and female) Danish study demonstrated that 51% of participants regarded their sexual life as a failure or as dysfunctional; however, 45% reported being satisfied with their sexual life, with these individuals being more likely to have partners. Fecal incontinence was associated with poorer sexual function and less satisfaction; however, urinary incontinence was not. This study also found there were long-lasting effects of sex education provided in teenage years because nearly 50% of the participants reported that it was useful.[112]

For females with MMC, menstruation and fertility are thought to be normal, and affected individuals are capable of becoming pregnant.[117] Pregnancy complications can be common, including recurrent urinary tract infection, worsening kyphoscoliosis, VP shunt malfunction, and failure of genitourinary diversions. Vaginal delivery is generally indicated, especially in those with a VP shunt, and cesarean section should be performed only for obstetric reasons such as fetal distress, failure to progress labor, or inadequacy of the birth canal.[89]

Late Neurologic Considerations

Adults with MMC remain at risk for developing neurologic complications of their disease as well as other secondary conditions. Complications can include VP shunt infections and malfunctions, syringomyelia, and symptomatic type II Chiari malformation.[24] Adolescents and adults with MMC may remain shunt dependent throughout their lifetime. However, some may “outgrow” the need for shunting, and fractured shunt tubing may be an incidental finding on unrelated radiologic studies.[32] Up to 10% of adults with VP shunts can develop chronic idiopathic headaches requiring specialized management. However, at presentation, new headaches should always receive a comprehensive evaluation to rule out treatable and potentially life-threatening causes.[35] Peripheral nerve entrapment syndromes, overuse syndromes, and herniated discs need to be considered in the differential diagnosis of new upper limb symptoms in adults more so than in children. Seizures ranging in incidence from 3% to 23% in the adult population have also been reported and are thought to be multifactorial in origin.[110] Adults with MMC can also develop syringomyelia, presenting with complaints of pain, paresthesia, and weakness in the upper limbs, or tethering of the spinal cord, presenting with new weakness, spasticity, or changes in bowel and bladder function. As in children, both conditions often require surgical intervention.[85]

Postsecondary Education, Vocational, and Independent Living Considerations

Most individuals with MMC complete high school, and approximately 50% move on to further education. However, little is published regarding the educational levels achieved, employment status, or living situation of adults with MMC. Bowman et al., in a long-term follow-up survey of 71 individuals, reported that 85% of individuals who survived to adulthood either attended or graduated from high school and/or college, with 36% requiring special education services; 45% of participants were employed, and 15% lived independently.[18] College graduation rate was reported to be only 14.6% in another small study.[51] A Canadian study of a mixed population of cerebral palsy and MMC found that transportation was a major barrier to employment, fewer females were employed than males, and employment rate was inversely related to IQ.[71] Other studies have reported rates of employment ranging from 25% to 62.5%, depending on the criteria being considered, including intelligence, academic qualifications, behavior, continence, and severity of physical disability. Rates of independent living have similarly been reported with wide variability ranging from 14% to 41%.[71,109] This variability clearly indicates the need for additional research in this area.

Life Expectancy

Individuals with MMC face increased chance of death throughout their lifespan compared to the general population.[90] Prior to the 1970s with the advent of cerebrospinal shunting, life expectancy was very short. A study of children born between 1963 and 1971 found one in three of the cohort (40/117) died before the age of 5 years.[79] The 1-year mortality rate for individuals with MMC in the 1970s was 25%, with the probability of survival at 75%. A successive improvement in 1-year mortality was seen in those born in the 1980s (14%) and in 2000 to 2010 (6%). In those born between 2010 and 2020, there was no further improvement.[7]

While the most common causes of childhood deaths were congenital abnormalities, hydrocephalus, and infections, adult deaths were most commonly due to hydrocephalus, infections, substance abuse, and self-inflicted injuries. Self-inflicted injuries, whether suicidal or of unclear intent, were more common than in the general population.[7] Many of these causes are potentially preventable and, again, argue in favor of multidisciplinary, comprehensive care systems for adults with MMC.

Summary

MMC and other spinal dysraphisms are prevalent around the world. Prevention strategies, including folic acid supplementation, can have a profound impact on the prevalence and severity and should be implemented; however, even with folic acid supplementation, NTDs continue to occur. Early detection and diagnosis provide families with options regarding fetal surgery and delivery as well as education and expectations for lifetime care. A multidisciplinary team should be present to address the many issues associated with this diagnosis, including initial surgical repair of the spinal defect, latex allergy, hydrocephalus and Chiari II malformation, tethered cord syndrome and shunt malfunction, orthopedic concerns regarding the spine and lower limbs, neurogenic bowel and bladder, potential skin injury, psychosocial and cognitive impairments, and endocrine abnormalities. Careful education and therapeutic intervention should be instituted to obtain the best functional outcomes. Additional information for providers and families can be found through the Spina Bifida Association (www.sbaa.org).

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